

Technology Stack

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit	Enables a rapid, all-Python interactive UI (input form, results, live dashboard) without building a separate JS frontend.
2	Backend / Server-Side	Python 3.11 (modular: emotion_detection, ai_response, logging_system, analytics_dashboard)	Tight integration with the TensorFlow/PyTorch ML libraries and Streamlit's Python-native execution model.
3	Database / Data Storage	CSV file storage (pandas)	Simple, transparent logging of every interaction as a row, with no separate database server needed for a prototype of this scale.
4	Cloud / Hosting / Deployment	Local machine (venv) / Streamlit Community Cloud	Straightforward, low-cost way to run and demo a Streamlit app without managing dedicated servers.
5	Version Control & CI/CD	Git & GitHub	Standard collaborative version control for the 3-member team, with room to add GitHub Actions checks later.
6	Third-Party APIs / Other Tools	Google Gemini API	Generates natural, personalized guidance text tailored to each detected emotion and confidence level.
7	ML Model Frameworks	TensorFlow / Keras (BiLSTM) & PyTorch / Transformers (BERT)	Runs two independent emotion-classification architectures so predictions can be cross-checked for reliability.